A differential equation is an equation with a function and one or more of its derivatives. (i.e) an equation with the function $y = f(x)$ and its derivatives $\frac{dy}{dx}, \frac{d^2y}{dx^2}$ … is called differential equation.

For example,

(i) $\frac{dy}{dx} = x + 5$ (ii) $\frac{dy}{dx} = \frac{\sqrt{1-x^2}}{\sqrt{2-y^2}}$

(iii) $\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 = 0$ (iv) $\frac{d^2x}{dt^2} + m^2x = 0$

Differential equations are of two types. One is ordinary differential equations and other one partial differential equations. Here we study only ordinary differential equations.

4.1 Formation of ordinary differential equations

4.1.1 Definition of ordinary differential equation

An ordinary differential equation is an equation that involves some ordinary derivatives $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots\right)$ of a function $y = f(x)$. Here we have one independent variable.

4.1.2 Order and degree of a differential equation

The highest order derivative present in the differential equation is the order of the differential equation.

Degree is the highest power of the highest order derivative in the differential equation, after the equation has been cleared from fractions and the radicals as for as the derivatives are concerned.

For example, consider the differential equation

$$\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^5 + y = 7$$

Here the highest order derivatives is $\frac{d^3y}{dx^3}$ (i.e 3rd order derivative). So the order of the differential equation is 3.

$\frac{d^3y}{dx^3}$ Now the power of highest order derivative is 1.

$\therefore$ The degree of the differential equation is 1.

Example 4.1

Find the order and degree of the following differential equations.

(i) $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 + 4y = 0$

(ii) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 3y = 0$

(iii) $\frac{d^3y}{dx^3} - 3\left(\frac{dy}{dx}\right)^6 + 2y = x^2$

(iv) $\left[1 + \frac{d^2y}{dx^2}\right]^{\frac{3}{2}} = a\frac{d^2y}{dx^2}$

(v) $y' + (y'')^2 = (x + y'')^2$

(vi) $\frac{d^3y}{dx^3} - \left(\frac{dy}{dx}\right)^{\frac{1}{2}} = 0$

(vii) $y = 2\left(\frac{dy}{dx}\right)^2 + 4x\frac{dx}{dy}$

Solution

(i) $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 + 4y = 0$

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